Introduction to Time Series Forecasting

01

Outline

- (20) Introduction
  - What is time series data?
  - Notation
- (30) Benchmark/baseline forecasting methods
  1) Naïve; 2) Seasonal naïve; 3) Mean; 4) Drift
- (20) Model evaluation
- (30) Course roadmap: Three forecasting families
  - Group I Baseline Group II Statistical models Group III Machine-learning models
  - Excel vs R
- (20) Forecasting objectives and uncertainty
  - Stock quantity and service level requirement

Two common types of data: cross-sectional vs time series

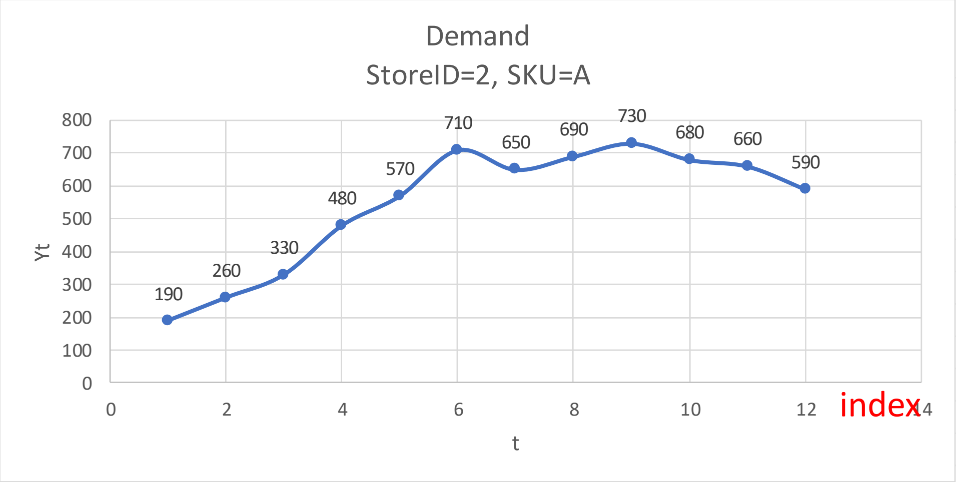
- **Cross-sectional**: Snapshot at a single point in time, comparing different items (e.g., salaries of baseball players in 2024)

Years_in_Majors	Career_ERA	Innings_Pitched	Career_Wins	Career_Losses	Salary_K
2	4.97	23.1	1	2	62.5
14	3.51	55.2	83	68	62.5
2	4.48	105.1	7	8	62.5
1	5.05	87.1	7	4	70
2	4.27	91.1	5	7	70
1	4.61	121	8	9	75
1	2.89	37.1	3	4	75
1	5.31	120.1	4	7	75
2	4.66	82	5	7	77.5

Row independent

- Time series: A **time series** can be thought of as a list of numbers (the **measurements**), along with some information about what times those numbers were recorded (the **index**).

t	Date	Demand StoreID=2, SKU=A
1	2001-01-01	190
2	2001-04-01	260
3	2001-07-01	330
4	2001-10-01	480
5	2002-01-01	570
6	2002-04-01	710
7	2002-07-01	650
8	2002-10-01	690
9	2003-01-01	730
10	2003-04-01	680
11	2003-07-01	660
12	2003-10-01	590

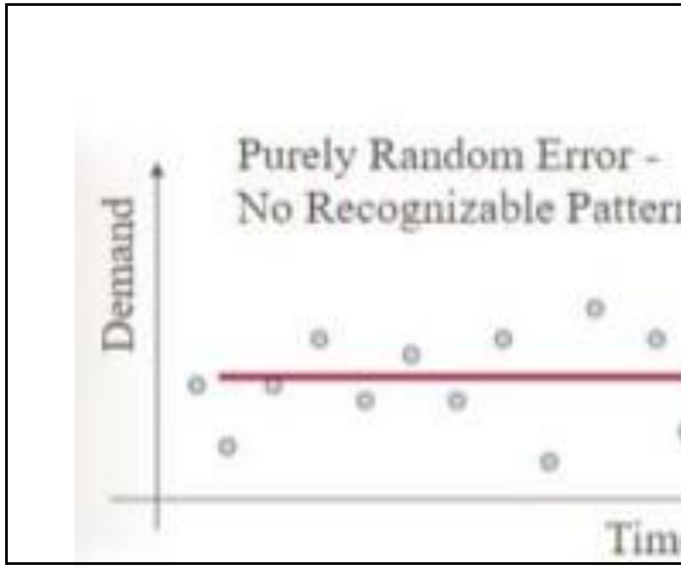


- In the above time series, index represents a) days; b) weeks; c) months; d) quarters; or e) years?

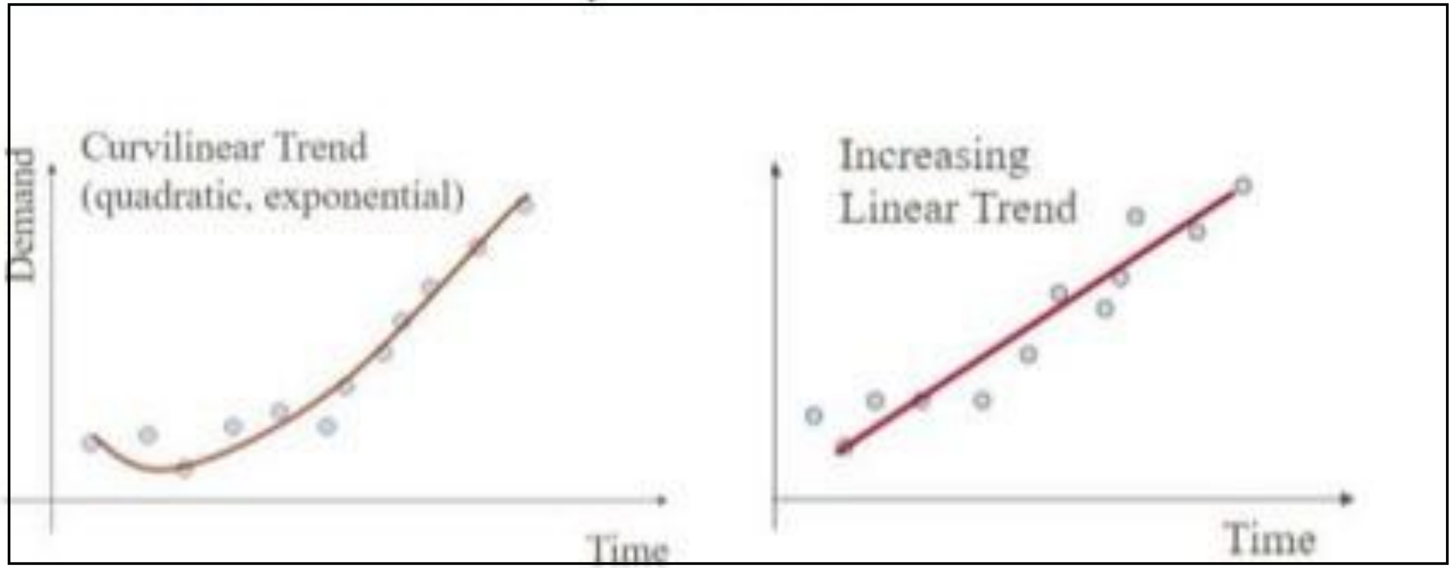
Examples: Which pattern?

- School uniform sales over months
- Average household incomes over years
- Daily stock closing price
- Position of drunk person walking on a straight line

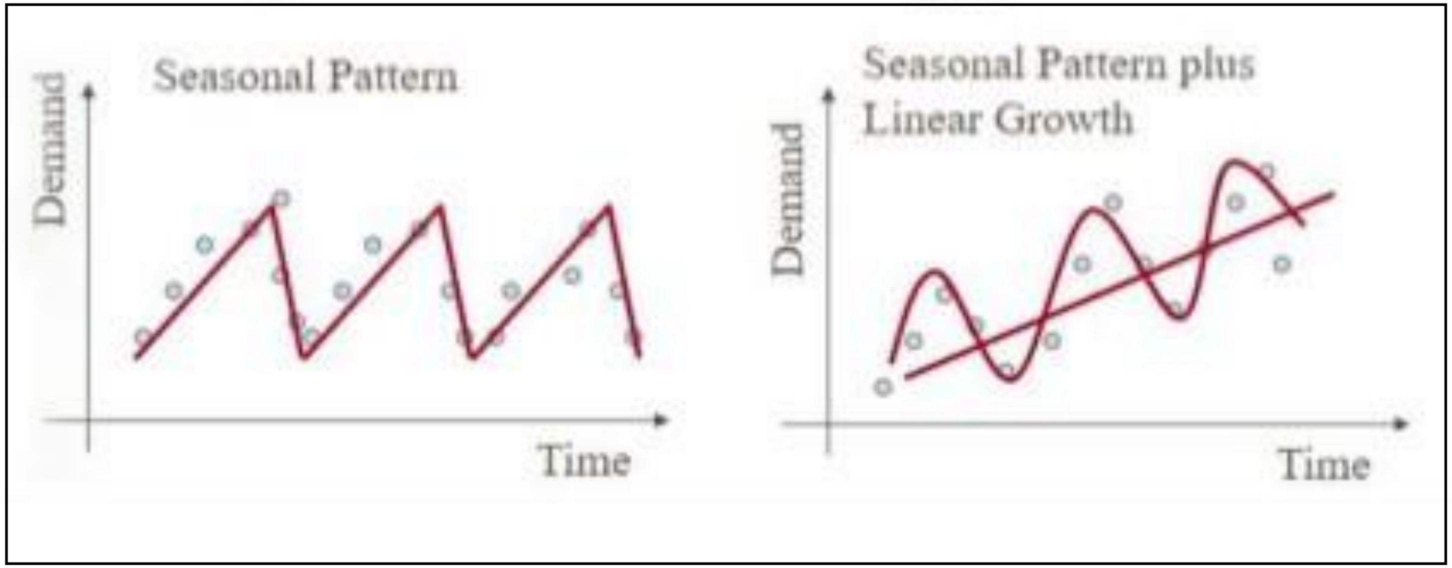
Common time series patterns



Level



Trend



Seasonal

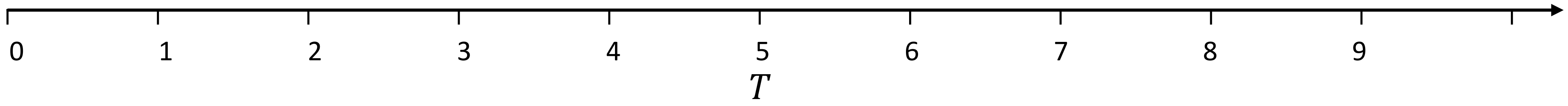
Notation

Notation for univariate time series

$\hat{y}_{6|5}$
$\hat{y}_{8|5}$
$\hat{y}_{8|7}$

The h-step forecast (estimate of y_{T+h}) based on data $y_1, \ldots, y_T$

$\hat{y}_{T+h|T}$

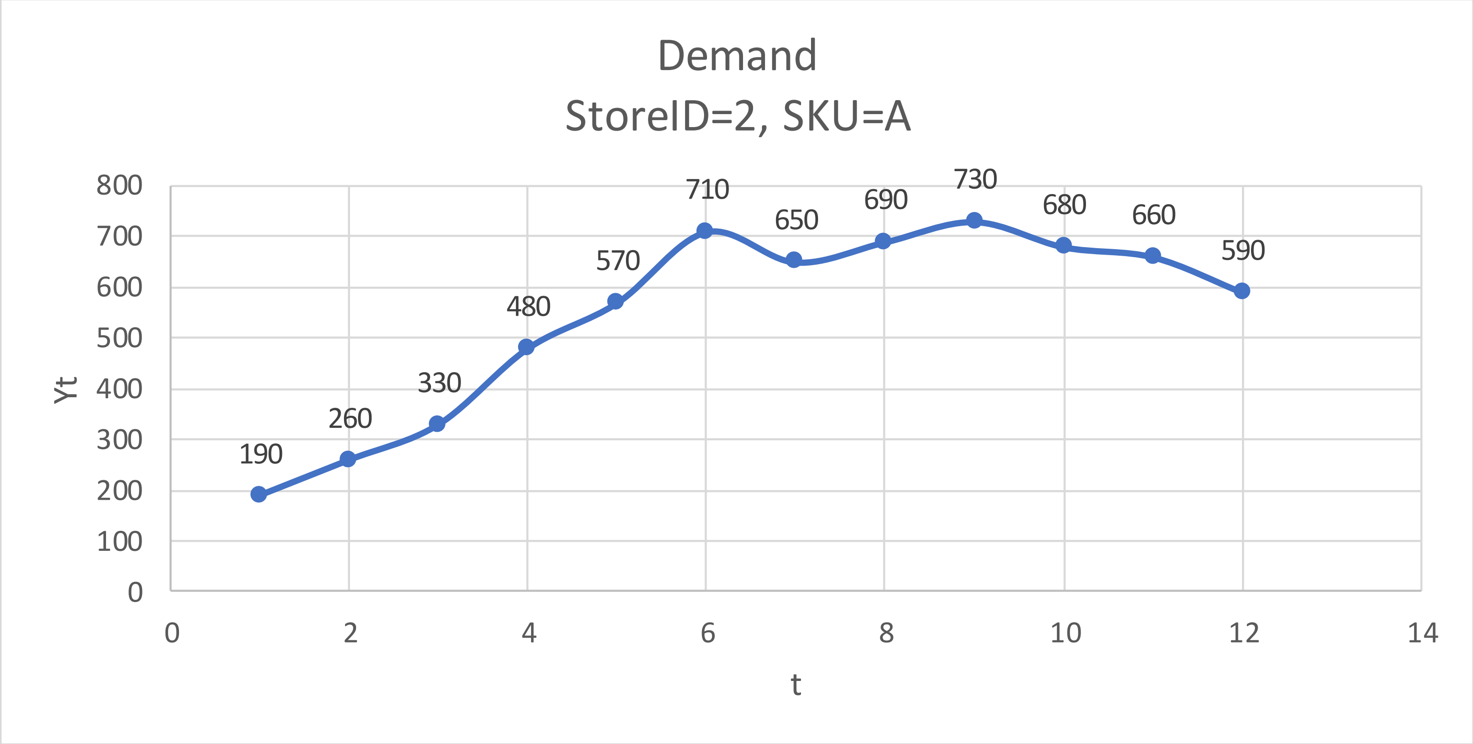


Small Example: **Time plot**

StoreID =2, SKU =A

File `02-introTS-FC`
Sheet `2A`

t	Date	Demand StoreID=2, SKU=A
1	2001-01-01	190
2	2001-04-01	260
3	2001-07-01	330
4	2001-10-01	480
5	2002-01-01	570
6	2002-04-01	710
7	2002-07-01	650
8	2002-10-01	690
9	2003-01-01	730
10	2003-04-01	680
11	2003-07-01	660
12	2003-10-01	590



Time index based on trading days observed, rather than calendar days

Example: Google's daily closing stock price

In Figure 5.8, the non-seasonal methods are applied to Google's daily closing stock price in 2015, and used to forecast one month ahead. Because stock prices are not observed every day, we first set up a new time index based on the trading days rather than calendar days.

```
# Re-index based on trading days
google_stock <- gafa_stock |>
  filter(Symbol == "GOOG", year(Date) >= 2015) |>
  mutate(day = row_number()) |>
  update_tsibble(index = day, regular = TRUE)
# Filter the year of interest
google_2015 <- google_stock |> filter(year(Date) == 2015)
```

Standard TS methods assume equal spacing

OriginalDate	ChargeableWeight
2-Apr-2019	6886
3-Apr-2019	3612.5
4-Apr-2019	6620.5
5-Apr-2019	4960.5
6-Apr-2019	10037
7-Apr-2019	2876
8-Apr-2019	2318
9-Apr-2019	1689.5
10-Apr-2019	18030
11-Apr-2019	5313.5
12-Apr-2019	7340
13-Apr-2019	8138
14-Apr-2019	5087
17-Apr-2019	11.5
18-Apr-2019	1485.5
19-Apr-2019	12607.5
20-Apr-2019	10925
21-Apr-2019	3964.5
22-Apr-2019	7636
23-Apr-2019	2478
24-Apr-2019	11216.5
25-Apr-2019	3708
26-Apr-2019	2914.5
27-Apr-2019	286.5
30-Apr-2019	488.5

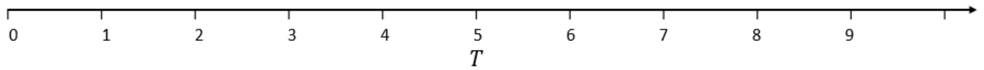
Original transaction SAP data from IS projects

Because this company operates every day, missing dates should be added with chargeable weight = 0.

OriginalDate	ChargeableWeight
1-Apr-2019	0
2-Apr-2019	6886
3-Apr-2019	3612.5
4-Apr-2019	6620.5
5-Apr-2019	4960.5
6-Apr-2019	10037
7-Apr-2019	2876
8-Apr-2019	2318
9-Apr-2019	1689.5
10-Apr-2019	18030
11-Apr-2019	5313.5
12-Apr-2019	7340
13-Apr-2019	8138
14-Apr-2019	5087
15-Apr-2019	0
16-Apr-2019	0
17-Apr-2019	11.5
18-Apr-2019	1485.5
19-Apr-2019	12607.5
20-Apr-2019	10925
21-Apr-2019	3964.5
22-Apr-2019	7636
23-Apr-2019	2478
24-Apr-2019	11216.5
25-Apr-2019	3708
26-Apr-2019	2914.5
27-Apr-2019	286.5
28-Apr-2019	0
29-Apr-2019	0

Daily time-series data

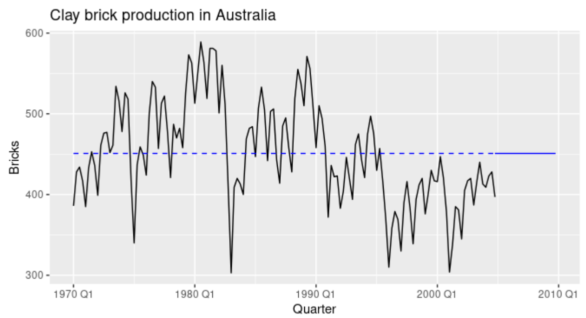
Benchmark forecasting methods



Simple Forecasting Methods

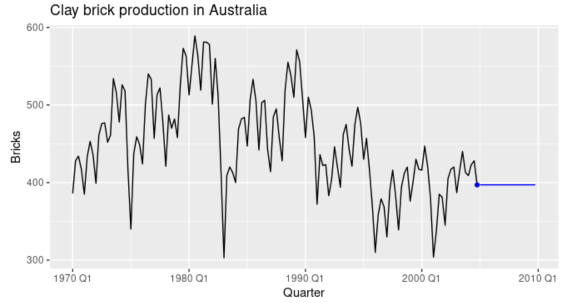
- Mean method

$$\hat{y}_{T+h|T} = \frac{1}{T}(y_1 + \cdots + y_T)$$



- Naïve method

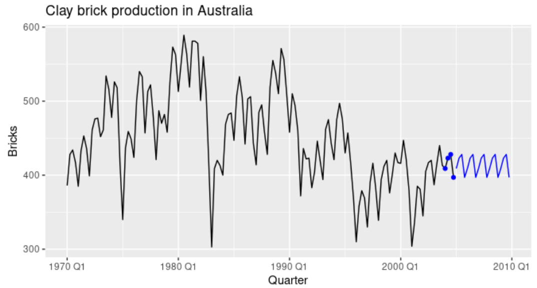
$$\hat{y}_{T+h|T} = y_T$$



- Seasonal naïve method

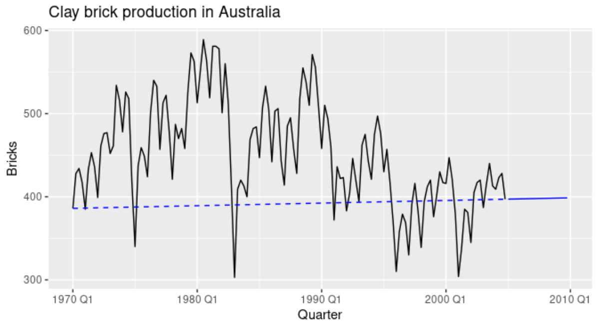
$$\hat{y}_{T+h|T} = y_{T+h-m(k+1)},$$

where m = the **seasonal period**, and k is the integer part of $(h-1)/m$



- Drift method

$$\hat{y}_{T+h|T} = y_T + \underbrace{\frac{h}{T-1}\sum_{t=2}^{T}(y_t - y_{t-1})}_{\text{Average change}} = y_T + h\underbrace{\left(\frac{y_T - y_1}{T-1}\right)}_{\text{slope}}$$



Data pattern: Methods
Level: Mean or Naïve or Seasonal Naïve or Drift?
Trend:
Seasonal:

File `02-introTS-FC`
Sheet `2A`

Small Example
StoreID =2, SKU =A

$$\hat{y}_{T+h|T} = y_T \qquad \hat{y}_{T+h|T} = y_{T+h-m(k+1)}$$

$$\hat{y}_{T+h|T} = (y_1 + \cdots + y_T)/T$$

Set	t	Date	Demand	FC.Mean	FC.Naive	FC.Seasonal Naïve
Train	1	2001-01-01	y_1 =190			
Train	2	2001-04-01	260			
Train	3	2001-07-01	330			
Train	4	2001-10-01	480			
Train	5	2002-01-01	570			
Train	6	2002-04-01	710			
Train	7	2002-07-01	650			
Train	T=8	2002-10-01	y_8 =690			
Test	9	2003-01-01	730	485	690	570
Test	10	2003-04-01	680	485	690	710
Test	11	2003-07-01	660	485	690	650
Test	12	2003-10-01	590	485	690	690

Quick check (not this train-test split)

- $\hat{y}_{13|8}$
- $\hat{y}_{13|12}$

Small Example

StoreID =2, SKU =A

File `02-introTS-FC`
Sheet `2A`

$$\hat{y}_{T+h|T} = \quad y_T + h\left(\frac{y_T - y_1}{T-1}\right)$$

Set	t	Date	Demand	FC.Mean	FC.Naive	FC.Seasonal Naïve	FC.Drift
							71.43
Train	1	2001-01-01	y_1 =190				
Train	2	2001-04-01	260				
Train	3	2001-07-01	330				
Train	4	2001-10-01	480				
Train	5	2002-01-01	570				
Train	6	2002-04-01	710				
Train	7	2002-07-01	650				
Train	T=8	2002-10-01	y_8 =690				
Test	9	2003-01-01	730	485	690	570	761
Test	10	2003-04-01	680	485	690	710	833
Test	11	2003-07-01	660	485	690	650	904
Test	12	2003-10-01	590	485	690	690	976

Model evaluation

Train and test split

Error measurements

Metrics in **bold** are commonly used.

Measures of Forecast Errors

- Forecast error in period t=1,…,T

  Error = Actual – Forecast

  $e_t = y_t - \hat{y}_t$

- Positive error: under-forecast (forecast too low)
- Negative error: over-forecast (forecast too high)

Forecast Error Measurement	
Mean Absolute Error	**MAE** = $(1/T) \sum_t \lvert e_t \rvert$
Mean Squared Error	MSE = $(1/T) \sum_t e_t^{\,2}$
Root Mean Squared Error	**RMSE**= $\sqrt{\text{MSE}}$
Mean Bias Error	MBE = $(1/T) \sum_t e_t$
Mean Percentage Error	MPE $= (1/T) \sum_t e_t / y_t$
Mean Absolute Percentage Error	$\mathbf{MAPE} = \left(\frac{1}{\text{T}}\right) \sum_{\text{t}} \lvert e_t \rvert / y_t$ $\text{MAPE (\%)} = (100/\text{T}) \sum \lvert e_t \rvert / y_t$
Symmetric MAPE Hyndman and Koehler (2006) recommend that the sMAPE not be used. It is included here only because it is sometimes used in practice.	$\text{SMAPE}_1 = \left(\frac{1}{T}\right) \sum \frac{\lvert e_t \rvert}{(\lvert A_t \rvert + \lvert F_t \rvert)/2}$ $\text{SMAPE}_2 = \left(\frac{100\%}{T}\right) \sum \frac{\lvert e_t \rvert}{(\lvert A_t \rvert + \lvert F_t \rvert)}$
Mean Absolute Scaled Error	**MASE** = $\frac{\text{MAE of model}}{\text{MAE of (seasonal) naive}} = \frac{\frac{1}{J} \sum_j \lvert e_j \rvert}{\frac{1}{T-1} \sum_{t=2}^{T} \lvert Y_t - Y_{t-1} \rvert}$
Theil's U	U = $\frac{\text{RMSE(model)}}{\text{RMSE(naive model)}}$

Small Example

StoreID =2, SKU =A

File `02-introTS-FC`
Sheet `2A`

$$\hat{y}_{T+h|T} = y_T + h\left(\frac{y_T - y_1}{T-1}\right)$$

$$\hat{y}_{T+h|T} = y_T \qquad \hat{y}_{T+h|T} = y_{T+h-m(k+1)}$$

$$\hat{y}_{T+h|T} = (y_1 + \cdots + y_T)/T$$

Set	t	Date	Demand	FC.Mean	FC.Naive	FC.Seasonal Naïve	FC.Drift 71.43
Train	1	2001-01-01	y_1 =190				
Train	2	2001-04-01	260				
Train	3	2001-07-01	330				
Train	4	2001-10-01	480				
Train	5	2002-01-01	570				
Train	6	2002-04-01	710				
Train	7	2002-07-01	650				
Train	T=8	2002-10-01	y_8 =690				
Test	9	2003-01-01	730	485	690	570	761
Test	10	2003-04-01	680	485	690	710	833
Test	11	2003-07-01	660	485	690	650	904
Test	12	2003-10-01	590	485	690	690	976
Test MAE				180	45	75	204

Which metric should be used?

- Scale-invariant measures allow comparison across different units.
  - MAPE
  - Normalized RMSE (NRMSE) and normalized NMAE normalize errors by a scale factor, such as the mean or range. Always state the scale factor used.
  - MASE < 1, better than (seasonal naïve)
- Robust measures are relatively insensitive to large forecast errors (outliers), e.g., MAE
  - RMSE not robust; penalizes large errors
- Bias detection: MBE

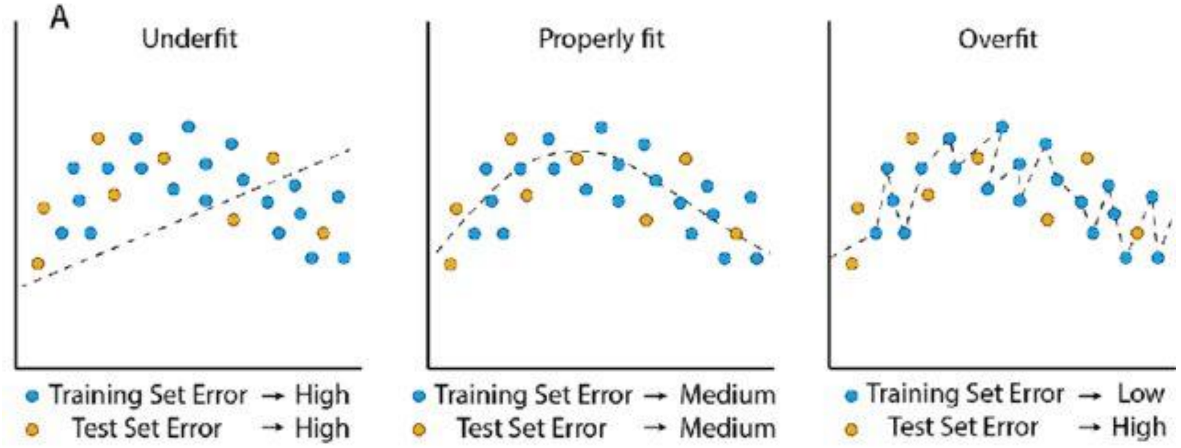
MAE = $(1/T) \sum_t \lvert e_t \rvert$
MSE = $(1/T) \sum_t e_t^{\,2}$
RMSE= $\sqrt{\mathrm{MSE}}$
MBE = $(1/T) \sum_t e_t$
MPE = $(1/T) \sum_t e_t / y_t$
$\mathbf{MAPE} = \left(\frac{1}{\mathrm{T}}\right) \sum_{\mathrm{t}} \lvert e_t \rvert / y_t$ $\mathrm{MAPE}\ (\%) = (100/\mathrm{T}) \sum \lvert e_t \rvert / y_t$
$\mathrm{SMAPE}_1 = \left(\frac{1}{T}\right) \sum \frac{\lvert e_t \rvert}{(\lvert A_t \rvert + \lvert F_t \rvert)/2}$ $\mathrm{SMAPE}_2 = \left(\frac{100\%}{T}\right) \sum \frac{\lvert e_t \rvert}{(\lvert A_t \rvert + \lvert F_t \rvert)}$
MASE = $\frac{\text{MAE of model}}{\text{MAE of (seasonal) naive}}$
U = $\frac{\text{RMSE(model)}}{\text{RMSE(naive model)}}$

Data Partitioning

Training (in-sample)	Test (out-of-sample / holdout)

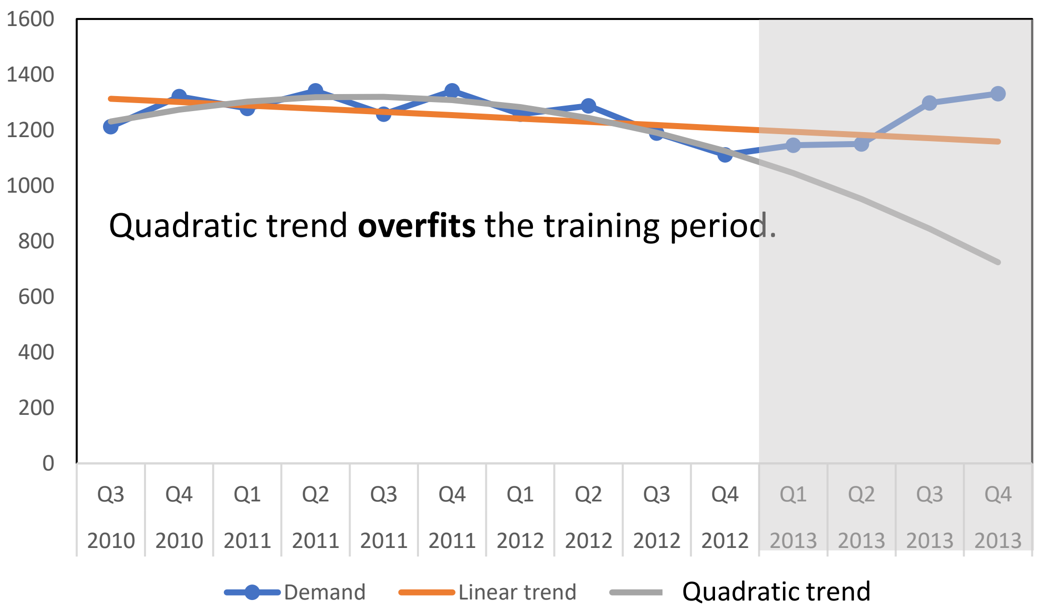
- Partitioning and testing for accuracy are standard practice in analytics.
- We use the **training data** to build the model and the **test data** to evaluate how well the model performs on unseen data.
- We do not use the test data when building, tuning, or preprocessing the model.
  - “Training” data: the partition we use to build the model,
  - “Test” data: the remainder of the data used only for final accuracy evaluation.
- This approach helps detect and reduce the risk of **overfitting**, where a model fits the training data well but performs poorly on new data.
- In time series forecasting, the split should respect time order.

Overfitting



	t	Year	Qtr	Demand	Linear trend	Quadratic trend
Train	1	2010	Q3	1212	1312.7	1231.2
Train	2	2010	Q4	1321	1300.9	1273.7
Train	3	2011	Q1	1278	1289.0	1302.6
Train	4	2011	Q2	1341	1277.2	1317.9
Train	5	2011	Q3	1257	1265.3	1319.7
Train	6	2011	Q4	1341	1253.5	1307.8
Train	7	2012	Q1	1257	1241.6	1282.4
Train	8	2012	Q2	1287	1229.8	1243.3
Train	9	2012	Q3	1189	1217.9	1190.7
Train	10	2012	Q4	1111	1206.1	1124.6
Test	11	2013	Q1	1145	1194.2	1044.8
Test	12	2013	Q2	1150	1182.3	951.4
Test	13	2013	Q3	1298	1170.5	844.5
Test	14	2013	Q4	1331	1158.6	724.0

Training	MAE	50.89	29.03
Test	MAE	95.35	339.83



Details later.

- Linear regression — Linear trend — $\hat{y}_t = 1324.6 - 11.9t$
- Polynomial regression — Quadratic trend (parabola) — $\hat{y}_t = 1175.2 + 62.9t - 6.8t^2$

Forecasting Models: A Roadmap for This Course

Model families (preview)

1. **Simple baselines**, e.g., mean; naïve; seasonal naïve; drift; moving average

- No model fitting, no parameters
- Serves as a minimum performance benchmark
- If a model cannot beat this, it should not be used

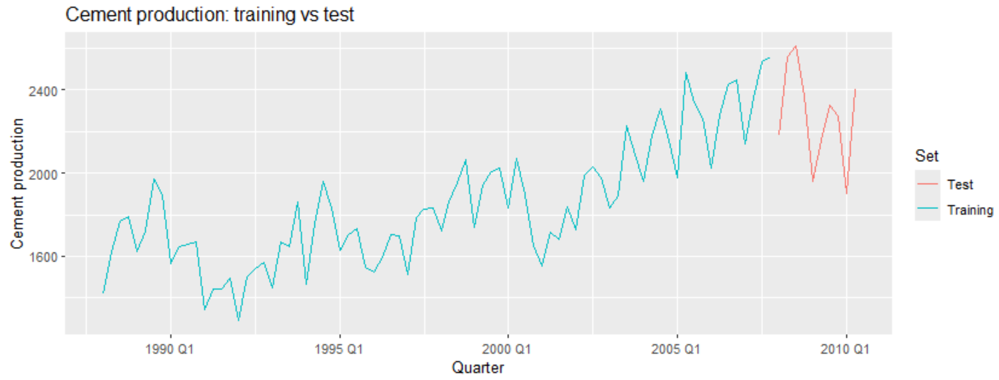
2. **Statistical time-series models**, e.g., ETS, ARIMA, regression, dynamic regression, dynamic harmonic regression

- Explicitly model trend, seasonality, and noise. Produce smooth and structured forecasts
- Well-suited for small to medium datasets
- Strong interpretability and clear assumptions

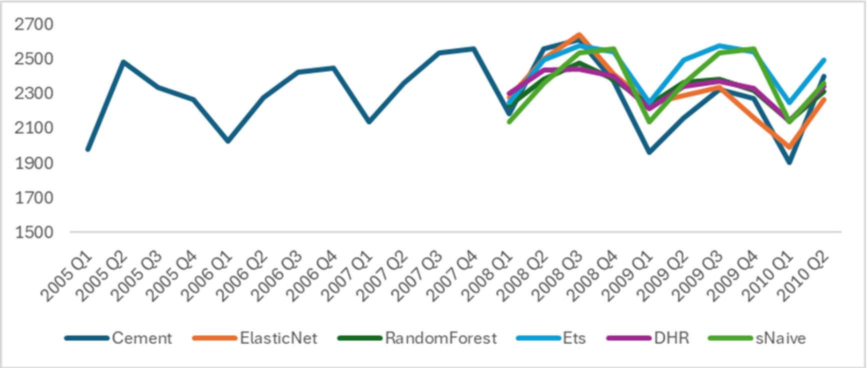
3. **Machine-learning forecasting**, e.g., Regularized regression (Elastic Net), Random Forest

- Treat forecasting as a supervised learning problem
- Depend on feature engineering (lags, averages, seasonality)
- More flexible, but higher risk of overfitting. Require careful cross-validation for time series

Example: Australian quarterly cement production



Train/Test split: Train 1998Q1-2007Q4 Test 2008Q1-2010Q2



Quarter	Cement	qtr_num	lag1	lag2	ma4	ma8	sin	cos
1988 Q1	1418	1				N/A	1.000	0.000
1988 Q2	1625	2	1418			N/A	0.000	-1.000
1988 Q3	1770	3	1625	1418		N/A	-1.000	0.000
1988 Q4	1791	4	1770	1625	1651.0	N/A	0.000	1.000
1989 Q1	1621	5	1791	1770	1701.8	N/A	1.000	0.000
1989 Q2	1719	6	1621	1791	1725.3	N/A	0.000	-1.000
1989 Q3	1972	7	1719	1621	1775.8	N/A	-1.000	0.000
1989 Q4	1894	8	1972	1719	1801.5	1726.3	0.000	1.000
1990 Q1	1565	9	1894	1972	1787.5	1744.6	1.000	0.000
1990 Q2	1645	10	1565	1894	1769.0	1747.1	0.000	-1.000
⋮								

Useful feature (preview)

- **Moving averages**
- MA4: Moving average of order 4
  - MA4 at time 4 = $\frac{y_4+y_3+y_2+y_1}{4}$ = 1651.0
  - MA4 at time 5 = $\frac{y_5+y_4+y_3+y_2}{4}$ = 1701.8
- MA8: Moving average of order 8
  - MA8 at time 10 = 8-period trailing moving average at time 10

Quarter	Cement	qtr_num	lag1	lag2	ma4	ma8	sin	cos
1988 Q1	1418	1				N/A	1.000	0.000
1988 Q2	1625	2	1418			N/A	0.000	-1.000
1988 Q3	1770	3	1625	1418		N/A	-1.000	0.000
1988 Q4	1791	4	1770	1625	1651.0	N/A	0.000	1.000
1989 Q1	1621	5	1791	1770	1701.8	N/A	1.000	0.000
1989 Q2	1719	6	1621	1791	1725.3	N/A	0.000	-1.000
1989 Q3	1972	7	1719	1621	1775.8	N/A	-1.000	0.000
1989 Q4	1894	8	1972	1719	1801.5	1726.3	0.000	1.000
1990 Q1	1565	9	1894	1972	1787.5	1744.6	1.000	0.000
1990 Q2	1645	10	1565	1894	1769.0	1747.1	0.000	-1.000
1990 Q3	1658	11	1645	1565	1690.5	1733.1	-1.000	0.000

- **Lag variables:** Lag-ℓ variable at time t, $B^{\ell} y_t \coloneqq y_{t-\ell}$
- Lag-1 variable
  - Lag-1 variable at time 2 = $By_2 = y_1$ = 1418
  - Lag-1 variable at time 3 = $By_3 = y_2$ = 1625
- Lag-2 variable
  - Lag-2 variable at time 3 := $B^2 y_3 = y_{3-2} = y_1$ = 1418
  - Lag-2 variable at time 4 := $B^2 y_4 = y_{4-2} = y_2 = 1625$
  - Lag-2 variable at time 10

Notation: Different books use different notation
- B backshift operator
- L lag operator

R

Creating Time Series Data in R

- Manual entry
  - Small examples and exam
- Read from file
  - When data are stored in Excel/CSV, use `read_csv()` from library `tidyverse`
- Use built-in R data sources
  - `fpp3` package, e.g., in your homework

```
demandvec <- c(30, 50, 30, 60, 10, 40,
               30, 30, 20, 40)
yFan <- tsibble(
  Year = 2001:2010,
  Demand = demandvec,
  index = Year
)

fc <- yFan %>%
  model(mMean = MEAN(Demand)) %>%
  forecast(h=1)

fc
```

```
df <- read_csv("smallExample.csv")
head(df)
tail(df)
glimpse(df)

myts <- df %>%
  mutate(Quarter = yearquarter(Date) ) %>%
  as_tsibble(key = c(StoreID, SKU ), index = Quarter   ) %>%
  select(-Date)

# Create training set
myts_train <- myts %>%
  filter(year(Quarter) <= 2002)

# Fit model on training set
myts_fit <- myts_train %>%
  model(
    Mean = MEAN(Demand),
    Naive = NAIVE(Demand),
    "Seasonal Naive" = SNAIVE(Demand),
    Drift = NAIVE(Demand ~ drift())
  )
myts_fit

accuracy(myts_fit)  # Residual on train

# Forecast on test set
myts_fc <- myts_fit %>%
  forecast(h=4)
myts_fc

accuracy(myts_fc, myts)  # Error on test
```

```
> aus_production
# A tsibble: 218 x 7 [1Q]
```

	Quarter	Beer	Tobacco	Bricks	Cement	Electricity	Gas
	<qtr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1956 Q1	284	5225	189	465	3923	5
2	1956 Q2	213	5178	204	532	4436	6
3	1956 Q3	227	5297	208	561	4806	7
4	1956 Q4	308	5681	197	570	4418	6

Excel vs R

```
df <- read_csv("smallExample.csv")
head(df)
tail(df)
glimpse(df)

myts <- df %>%
  mutate(Quarter = yearquarter(Date) ) %>%
  as_tsibble(key = c(StoreID, SKU ), index = Quarter   ) %>%
  select(-Date)

# Create training set
myts_train <- myts %>%
  filter(year(Quarter) <= 2002)

# Fit model on training set
myts_fit <- myts_train %>%
  model(
    Mean = MEAN(Demand),
    Naive = NAIVE(Demand),
    "Seasonal Naive" = SNAIVE(Demand),
    Drift = NAIVE(Demand ~ drift())
  )
myts_fit

accuracy(myts_fit)  # Residual on train

# Forecast on test set
myts_fc <- myts_fit %>%
  forecast(h=4)
myts_fc

accuracy(myts_fc, myts)  # Error on test
```

Set	t	Date	Demand	FC.Mean	FC.Naive	FC.Seasonal Naïve	FC.Drift
Train	1	2001-01-01	190				
Train	2	2001-04-01	260				
Train	3	2001-07-01	330				
Train	4	2001-10-01	480				
Train	5	2002-01-01	570				
Train	6	2002-04-01	710				
Train	7	2002-07-01	650				
Train	T=8	2002-10-01	690				
Test	9	2003-01-01	730	485	690	570	761
Test	10	2003-04-01	680	485	690	710	833
Test	11	2003-07-01	660	485	690	650	904
Test	12	2003-10-01	590	485	690	690	976
Test MAE				180	45	75	204

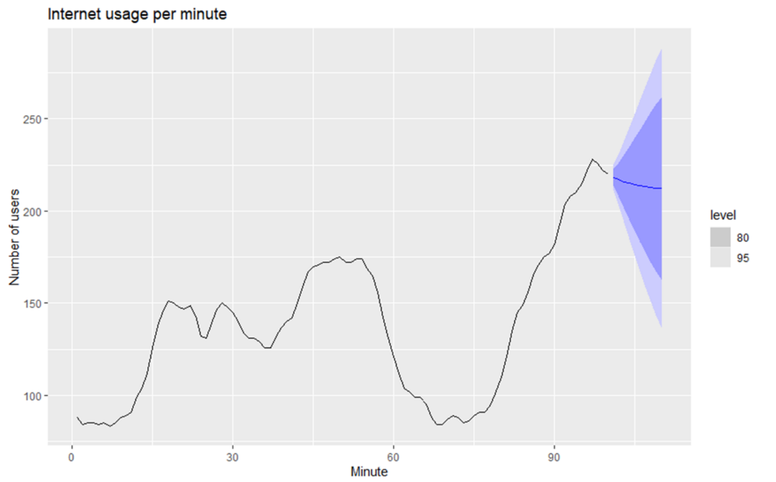
```
> myts_fc %>%
+   filter(StoreID == "2") %>%
+   filter(SKU == "A")
# A fable: 16 x 6 [1Q]
# Key:     StoreID, SKU, .model [4]
   StoreID SKU   .model         Quarter            Demand .mean
     <dbl> <chr> <chr>            <qtr>            <dist> <dbl>
 1       2 A     Mean           2003 Q1   N(485, 46414)   485
 2       2 A     Mean           2003 Q2   N(485, 46414)   485
 3       2 A     Mean           2003 Q3   N(485, 46414)   485
 4       2 A     Mean           2003 Q4   N(485, 46414)   485
 5       2 A     Naive          2003 Q1    N(690, 9314)   690
 6       2 A     Naive          2003 Q2   N(690, 18629)   690
 7       2 A     Naive          2003 Q3   N(690, 27943)   690
 8       2 A     Naive          2003 Q4   N(690, 37257)   690
 9       2 A     Seasonal Naive 2003 Q1  N(570, 123350)   570
10       2 A     Seasonal Naive 2003 Q2  N(710, 123350)   710
11       2 A     Seasonal Naive 2003 Q3  N(650, 123350)   650
12       2 A     Seasonal Naive 2003 Q4  N(690, 123350)   690
13       2 A     Drift          2003 Q1    N(761, 5616)   761.
14       2 A     Drift          2003 Q2   N(833, 12637)   833.
15       2 A     Drift          2003 Q3   N(904, 21061)   904.
16       2 A     Drift          2003 Q4   N(976, 30890)   976.

> accuracy(myts_fc, myts)  %>%
+   filter(StoreID == "2") %>%
+   filter(SKU == "A")
# A tibble: 4 x 12
  .model         StoreID SKU   .type     ME  RMSE   MAE    MPE  MAPE  MASE RMSSE   ACF1
  <chr>            <dbl> <chr> <chr>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl> <dbl>  <dbl>
1 Drift                2 A     Test  -204.  241.   204. -32.3   32.3 0.599 0.687  0.210
2 Mean                 2 A     Test   180   187.   180   26.6   26.6 0.529 0.532  0.126
3 Naive                2 A     Test   -25    56.1   45   -4.37   7.11 0.132 0.160  0.126
4 Seasonal Naive       2 A     Test    10    95.7   75    0.518 11.2  0.221 0.272 -0.166
```

Forecasting objectives & uncertainty

Distributional forecast

```
> myts_fc %>%
+   filter(StoreID == "2") %>%
+   filter(SKU == "A")
# A fable: 16 x 6 [1Q]
# Key:     StoreID, SKU, .model [4]
   StoreID SKU   .model          Quarter          Demand .mean
     <dbl> <chr> <chr>             <qtr>          <dist> <dbl>
 1       2 A     Mean            2003 Q1  N(485, 46414)   485
 2       2 A     Mean            2003 Q2  N(485, 46414)   485
 3       2 A     Mean            2003 Q3  N(485, 46414)   485
 4       2 A     Mean            2003 Q4  N(485, 46414)   485
 5       2 A     Naive           2003 Q1   N(690, 9314)   690
 6       2 A     Naive           2003 Q2  N(690, 18629)   690
 7       2 A     Naive           2003 Q3  N(690, 27943)   690
 8       2 A     Naive           2003 Q4  N(690, 37257)   690
 9       2 A     Seasonal Naive  2003 Q1 N(570, 123350)   570
10       2 A     Seasonal Naive  2003 Q2 N(710, 123350)   710
11       2 A     Seasonal Naive  2003 Q3 N(650, 123350)   650
12       2 A     Seasonal Naive  2003 Q4 N(690, 123350)   690
13       2 A     Drift           2003 Q1   N(761, 5616)  761.
14       2 A     Drift           2003 Q2  N(833, 12637)  833.
15       2 A     Drift           2003 Q3  N(904, 21061)  904.
16       2 A     Drift           2003 Q4  N(976, 30890)  976.
```

Types of forecasts



- Point forecast $\hat{y}_{T+h|T}$
- Interval forecast $\hat{y}_{T+h|T} \pm 1.96\hat{\sigma}_h$
- Distributional forecast $y_{T+h|T} \sim N(\hat{y}_{T+h|T}, \hat{\sigma}_h^2)$

Random variable $y_{T+h|T} = \hat{y}_{T+h|T} + e_{T+h}$

Assumption $e_{T+h} \sim N(0, \hat{\sigma}_h^2)$.

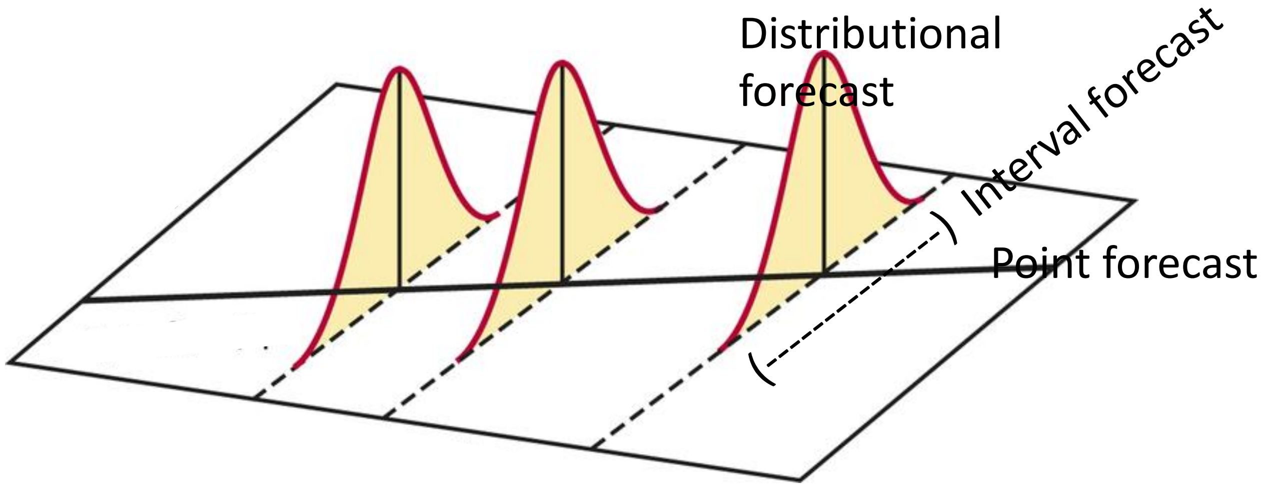
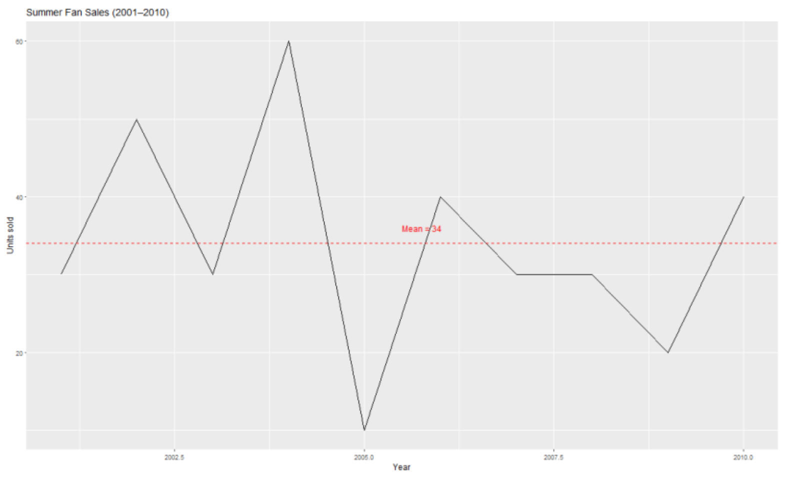


Figure 5.15: Five simulated future sample paths of the Google closing stock price based on a naïve method with bootstrapped residuals.

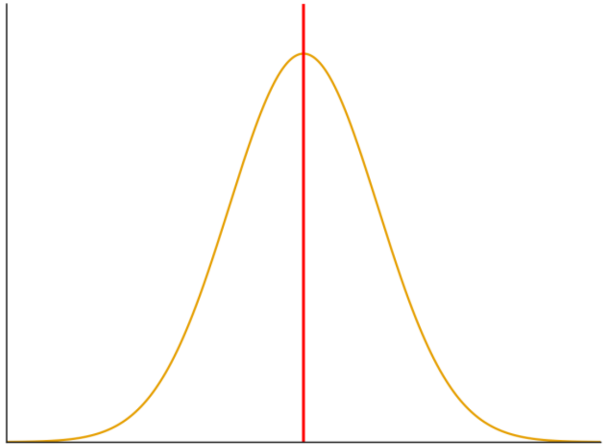
"Point forecast is almost always wrong!"

- Irwin's sells a particular model of a fan, with most of the sales being made in the summer months.
- The following is the number of sales of fans during the past 10 summers (2001-2010):

  30, 50, 30, 60, 10, 40, 30, 30, 20, 40.



- Assume that the yearly demand D follows the normal distribution with mean 34.
- If the stock quantity equals mean (34 fans), then the stockout probability is 50%, i.e., the service level is 50%.
- If we want a service level of 90%, what should the stock quantity be?
- Knowing only the mean is not enough! We need the **distribution** of the forecast.



Stock quantity S based on service level $100\alpha\%$

- Assume that demand follows normal distribution:
  $$D \sim N(\mu, \sigma^2)$$
- We want to find the smallest S such that
  $$\Pr(D \leq S) \geq \alpha$$
- Then, the solution is the α quantile (100α-th percentile)
  $$S = F^{-1}(\alpha; \mu, \sigma^2)$$
- $F^{-1}(t; \mu, \sigma^2)$ denotes the quantile function of $N(\mu, \sigma^2)$

```r
demandvec <- c(30, 50, 30, 60, 10, 40,
               30, 30, 20, 40)
yFan <- tsibble(
  Year = 2001:2010,
  Demand = demandvec,
  index = Year
)

fc <- yFan %>%
  model(mMean = MEAN(Demand)) %>%
  forecast(h=1)

fc

fc %>%
  as_tibble() %>%
  transmute(
    Year,
    q90 = quantile(Demand, 0.90),
    q95 = quantile(Demand, 0.95),
    q99 = quantile(Demand, 0.99)
  )
```

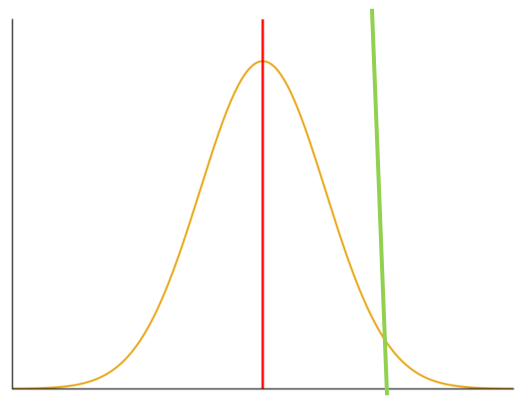
Suppose that the selected method is the mean method. How to select? MAE or something else? Later.

.model	Year	Demand	.mean
<chr>	<dbl>	<dist>	<dbl>
mMean	2011	N(34, 225)	34

$N(\mu, \sigma^2)$

Distributional/Probabilistic Forecast

Year	q90	q95	q99
<dbl>	<dbl>	<dbl>	<dbl>
2011	53.2	58.7	68.9



Given $\alpha = 0.90$, $S = F^{-1}(0.90; \mu = 34, \sigma^2 = 225)$ = 53.2.
Excel: `NORM.INV(0.9, 34, SQRT(225))`
Safety stock is $S - \mu = 53.2 - 34 = 19.2$

Hw

```
> aus_production
# A tsibble: 218 x 7 [1Q]
  Quarter  Beer Tobacco Bricks Cement Electricity   Gas
    <qtr> <dbl>   <dbl>  <dbl>  <dbl>       <dbl> <dbl>
1 1956 Q1   284    5225    189    465        3923     5
2 1956 Q2   213    5178    204    532        4436     6
3 1956 Q3   227    5297    208    561        4806     7
4 1956 Q4   308    5681    197    570        4418     6
```

- Dataset `aus_production`. Quarterly estimates of selected indicators of manufacturing production in Australia.
- Train/test split Training data 1988 Q1 – 2007 Q4; Test data 2008 Q1- 2010 Q2
- Consider the following methods

Method	R command
Mean	`MEAN(ColumnName)`
Naïve	`NAIVE(ColumnName)`
Seasonal naïve	`SNAIVE(ColumnName)`
Drift	`NAIVE(ColumnName ~ drift())`
ETS (later in the course)	`ETS(ColumnName)`
ARIMA (later in the course)	`ARIMA(ColumnName)`

Partial code

```
my_cement <- aus_production %>%
  filter(year(Quarter) >= 1988)

tail(my_cement)

train <- my_cement %>%
  filter(year(Quarter) <= 2007)

tail(train)

fit <- train %>%
  model(
    Mean = MEAN(Cement),
    Naive = NAIVE(Cement),
```

- Evaluate the methods and select the best one based on test RMSE.
- Do it in Excel for the first four methods, i.e., mean, naïve, seasonal naïve, drift

File `01-introTS-FC`
Sheet `aus_production`

Quarter	Beer	Tobacco	Bricks	Cement	Electricity	Gas
1956 Q1	284	5225	189	465	3923	5
1956 Q2	213	5178	204	532	4436	6
1956 Q3	227	5297	208	561	4806	7
1956 Q4	308	5681	197	570	4418	6
1957 Q1	262	5577	187	529	4339	5
1957 Q2	228	5651	214	604	4811	7
1957 Q3	236	5317	227	603	5259	7
1957 Q4	320	6152	222	582	4735	6

Appendix

Multivariate time series*

Univariate

Quarter	Sales
3/1/1981	1020.2
6/1/1981	889.2
9/1/1981	795
12/1/1981	1003.9
3/1/1982	1057.7
6/1/1982	944.4
9/1/1982	778.5
12/1/1982	932.5
3/1/1983	996.5
6/1/1983	907.7
9/1/1983	735.1
12/1/1983	958.1
3/1/1984	1034.1
6/1/1984	992.8
9/1/1984	791.7
12/1/1984	914.2

Multivariate

Wide format

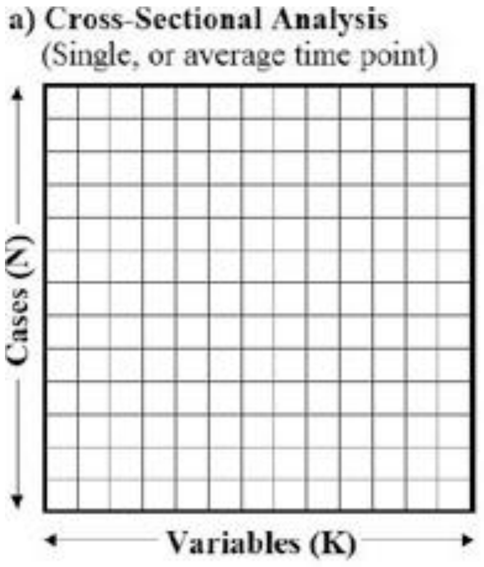
Quarter	Sales	AdBudget	GDP
3/1/1981	1020.2	659.2	251.8
6/1/1981	889.2	589	290.9
9/1/1981	795	512.5	290.8
12/1/1981	1003.9	614.1	292.4
3/1/1982	1057.7	647.2	279.1
6/1/1982	944.4	602	254
9/1/1982	778.5	530.7	295.6
12/1/1982	932.5	608.4	271.7
3/1/1983	996.5	637.9	259.6
6/1/1983	907.7	582.4	280.5

Long format

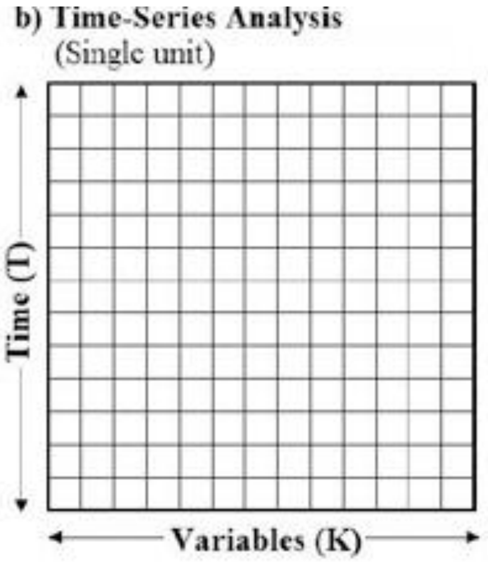
Each row has one observation

Quarter	Variable	Value
3/1/1981	Sales	1020.2
3/1/1981	AdBudget	659.2
3/1/1981	GDP	251.8
6/1/1981	Sales	889.2
6/1/1981	AdBudget	589
6/1/1981	GDP	290.9
	⋮	
6/1/1983	Sales	907.7
6/1/1983	AdBudget	582.4
6/1/1983	GDP	280.5

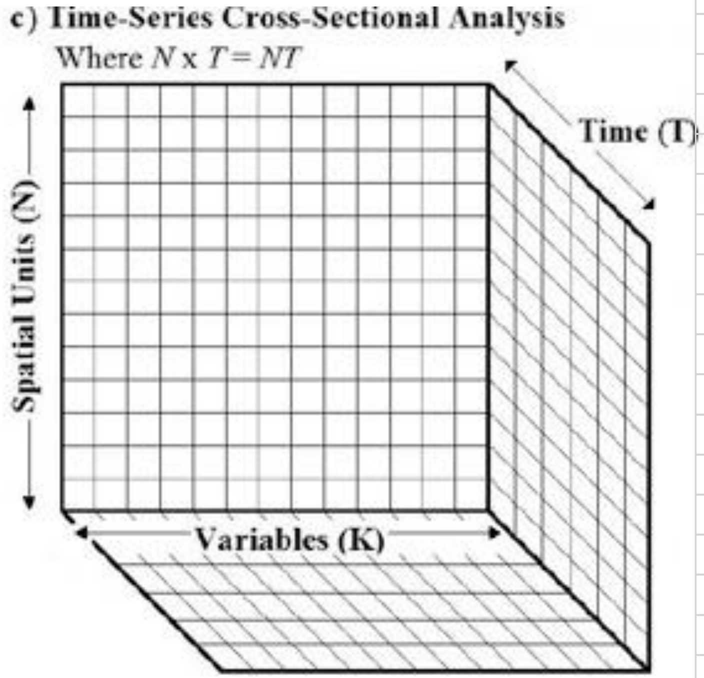
time-series cross-sectional, a.k.a., spatial time-series*



Year 2011		
ProvinceID	Population	Household
10	5674843	2459680
11	1203223	531987
12	5673560	539147



ProvinceID 10		
Year	Population	Household
2011	5674843	2459680
2012	5673560	2522855



Year 2012		
ProvinceID	Population	Household
10		
11		
12		

Year 2011		
ProvinceID	Population	Household
10	5674843	2459680
11	1203223	531987
12	5673560	539147

Key = ProvinceID & Type

YearEN	ProvinceID	Type	Observation
2011	10	Population	5674843
2011	11	Population	1203223
2011	12	Population	1122627
2012	10	Population	5673560
2012	11	Population	1223302
2012	12	Population	1141673
2011	10	Household	2459680
2011	11	Household	531987
2011	12	Household	539147
2012	10	Household	2522855
2012	11	Household	548883
2012	12	Household	561618
2011	10	Car	6849213
2012	10	Car	7523381
2011	11	Car	133610
2012	11	Car	135711
2011	12	Car	162734
2012	12	Car	163956
2011	10	Visitor	43763002
2012	10	Visitor	47185031
2011	11	Visitor	1489104
2012	11	Visitor	2107433
2011	12	Visitor	2004376
2012	12	Visitor	1663990

- Dataset with 10 years, 77 provinces, 3 variables
- How many rows?

Wide

A wide format contains values that do <u>not</u> repeat in the first column.

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

Quarter	Sales	AdBudget	GDP
3/1/1981	1020.2	659.2	251.8
6/1/1981	889.2	589	290.9
9/1/1981	795	512.5	290.8
12/1/1981	1003.9	614.1	292.4
3/1/1982	1057.7	647.2	279.1
6/1/1982	944.4	602	254
9/1/1982	778.5	530.7	295.6
12/1/1982	932.5	608.4	271.7
3/1/1983	996.5	637.9	259.6
6/1/1983	907.7	582.4	280.5

Long

A long format contains values that do repeat in the first column.

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
B	Points	91
B	Assists	17
B	Rebounds	28
C	Points	99
C	Assists	24
C	Rebounds	30
D	Points	94
D	Assists	28
D	Rebounds	31

Quarter	Variable	Value
3/1/1981	Sales	1020.2
3/1/1981	AdBudget	659.2
3/1/1981	GDP	251.8
6/1/1981	Sales	889.2
6/1/1981	AdBudget	589
6/1/1981	GDP	290.9
	⋮	
6/1/1983	Sales	907.7
6/1/1983	AdBudget	582.4
6/1/1983	GDP	280.5

Wide vs Long format*
Pivot vs Unpivot in Excel

Excel: Pivot
R: `pivot_wider()`

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

Advantages of wide format
- Easy to make quick comparison

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
B	Points	91
B	Assists	17
B	Rebounds	28
C	Points	99
C	Assists	24
C	Rebounds	30
D	Points	94
D	Assists	28
D	Rebounds	31

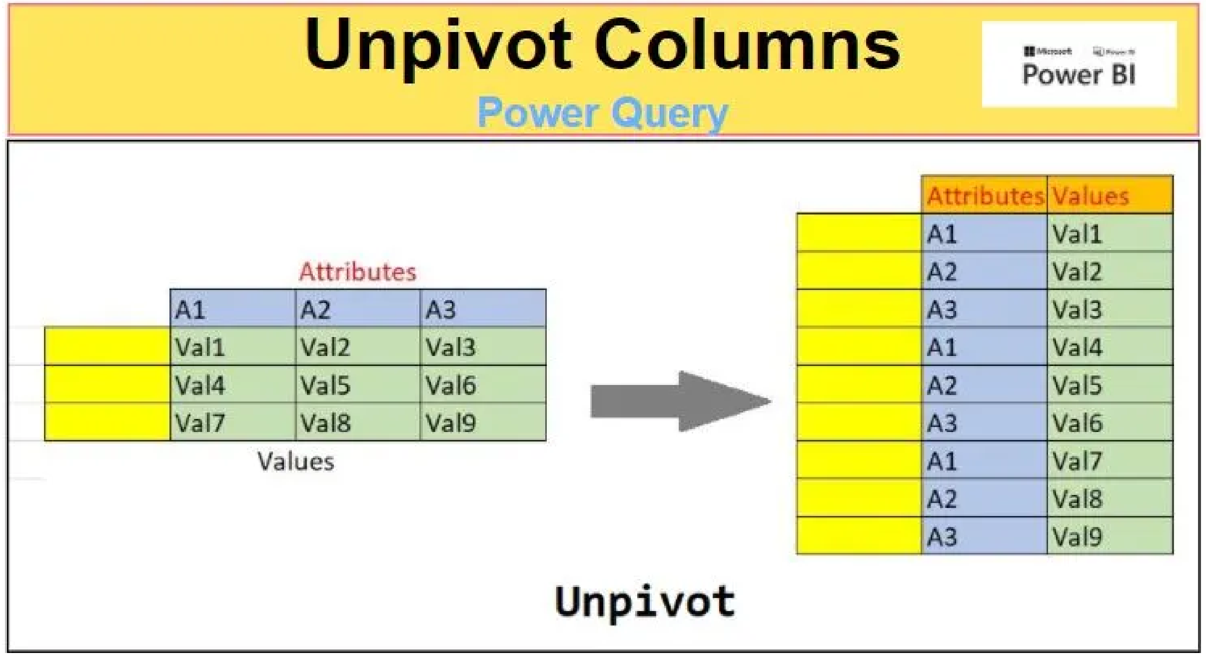
Excel: Unpivot
R: `pivot_longer()`

Advantage of long format
- Easy to store especially irregular and missing data points.
- Easy to transform to other types

Above, **key** = (Team, StatVariable)
What if key = (Team, StatVariable, Period)?
Period can be “First Half”, “Second Half”

```
d_wide <- data.frame(Team = c("A", "B", "C","D"),
                     Points = c(88, 91, 99, 94),
                     Assists = c(12, 17, 24, 28),
                     Rebounds = c(22, 28, 30, 31))

d_long <- pivot_longer(d_wide, !Team, names_to="Variable", values_to = "Value")
```

Excel: Unpivot Columns*



	1A	1B	2A	2B
1/1/2001	60	80	190	50
4/1/2001	160	30	260	20
7/1/2001	110	40	330	60
10/1/2001	80	20	480	40
1/1/2002	70	20	570	30
4/1/2002	140	10	710	60
7/1/2002	120	10	650	40
10/1/2002	60	20	690	20
1/1/2003	50	40	730	50
4/1/2003	120	30	680	40
7/1/2003	100	50	660	30
10/1/2003	50	60	590	60

Date	StoreID	SKU	Demanc
1/1/2001	1	A	60
4/1/2001	1	A	160
7/1/2001	1	A	110
10/1/2001	1	A	80
1/1/2002	1	A	70
4/1/2002	1	A	140
7/1/2002	1	A	120
10/1/2002	1	A	60
1/1/2003	1	A	50
4/1/2003	1	A	120
7/1/2003	1	A	100
10/1/2003	1	A	50
1/1/2001	1	B	80
4/1/2001	1	B	30
7/1/2001	1	B	40
10/1/2001	1	B	20
1/1/2002	1	B	20
4/1/2002	1	B	10
7/1/2002	1	B	10
10/1/2002	1	B	20
1/1/2003	1	B	40
4/1/2003	1	B	30
7/1/2003	1	B	50
10/1/2003	1	B	60
1/1/2001	2	A	190
4/1/2001	2	A	260
7/1/2001	2	A	330
10/1/2001	2	A	480
1/1/2002	2	A	570
4/1/2002	2	A	710
7/1/2002	2	A	650
10/1/2002	2	A	690
1/1/2003	2	A	730
4/1/2003	2	A	680
7/1/2003	2	A	660
10/1/2003	2	A	590
1/1/2001	2	B	50
4/1/2001	2	B	20
7/1/2001	2	B	60
10/1/2001	2	B	40
1/1/2002	2	B	30
4/1/2002	2	B	60
7/1/2002	2	B	40
10/1/2002	2	B	20
1/1/2003	2	B	50
4/1/2003	2	B	40
7/1/2003	2	B	30
10/1/2003	2	B	60

File: smallExample.csv